J1
SCK
CS_TC1
MISO
+5V
GND
TC1 MAX31855 (spring)

J2
SCK
CS_TC2
MISO
+5V
GND
TC2 MAX31855 (air)

J3
SCK
CS_TC3
MISO
+5V
GND
TC3 MAX31855 (heater)

J4
HX_DT
HX_SCK
+5V
GND
HX711 load cell

J5
SDA
SCL
+5V
GND
AS5600 angle

J6
PITOT_OUT
+5V
GND
MPXV7002 pitot

J7
FAN_TACH
FAN_PWM
+12V
GND
4-pin

J8
VIN_12V
GND
12V in

J9
SSR_DRV
+5V
Heater SSR input

R1
SDA
+5V
4k7

U1
Arduino_UNO
VIN
3V3
+5V
D0/RX
D1/TX
FAN_TACH
D2
BTN
D3
HX_DT
D4
HX_SCK
D5
SSR_PWM
D6
CS_TC3
D7
CS_TC2
D8
FAN_PWM_MCU
D9
CS_TC1
D10
D11
MISO
D12
SCK
D13
RESET
IOREF
A1
AREF
A0
PITOT_OUT
A1
LED_STATUS
A2
A3
SDA/A4
SDA
SCL/A5
SCL
SDA/A4
SDA
SCL/A5
SCL
GND
GND
GND
GND
GND
GND

R2
SCL
+5V
4k7

R3
FAN_TACH
+5V
10k

R4
FAN_PWM_MCU
FAN_PWM
1k

R5
SSR_PWM
SSR_GATE
100

R6
SSR_GATE
100k
GND

R7
LED_STATUS
LED_A
1k

C1
+5V
100n
GND

C2
+5V
100n
GND

C3
+5V
47u
GND

Q1
SSR_DRV
SSR_GATE
2N7000
GND

D1
VIN_12V
+12V
1N5822

D2
LED_A
GND
LED

SW1
BTN
START
GND

#PWR01
GND

#PWR02
+12V

Logic from USB, 12 V input feeds the fan only. SSR switches the halogen lamp (mains side off-board, with a KSD301 120 C thermal cut-out in series).

Sheet: /
File: sma_rig_shield.kicad_sch
Title: SMA brake duct bench rig — sensor shield
Size: A3 | Date: | **Rev: A**
KiCad E.D.A. 10.0.8 | Id: 1/1